

Auteur: Michael Livchits
 Studierichting: Informatica
 Oefeningenreeks 3G nr. 11.5

$$f(x) = x + \frac{25}{x}$$

VA: $x = 0$

$$\lim_{x \rightarrow 0^+} f(x) = \frac{25}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} f(x) = \frac{25}{0^-} = -\infty$$

SA:

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{x^2 + 25}{x^2} = \lim_{x \rightarrow \infty} 1 + \frac{25}{x^2} = 1$$

$$\lim_{x \rightarrow \infty} (f(x) - x) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + 25}{x} - x \right)$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 25}{x} \right) - \lim_{x \rightarrow \infty} (x) = \infty - \infty$$

$$\lim_{x \rightarrow \infty} \left(\frac{-x^2 + x^2 + 25}{x} \right) = \lim_{x \rightarrow \infty} \left(\frac{25}{x} \right) = 0$$

$$SA = x + 0$$

Ligging: $\begin{array}{c|c} x & 0 \\ \hline & - \quad / \quad + \end{array}$

$$x \rightarrow +\infty \Rightarrow f(x) > SA$$

$$x \rightarrow -\infty \Rightarrow f(x) < SA$$

$$f'(x) = 1 - \frac{25}{x^2} = \frac{x^2 - 25}{x^2}$$

Nulpunten: ± 5

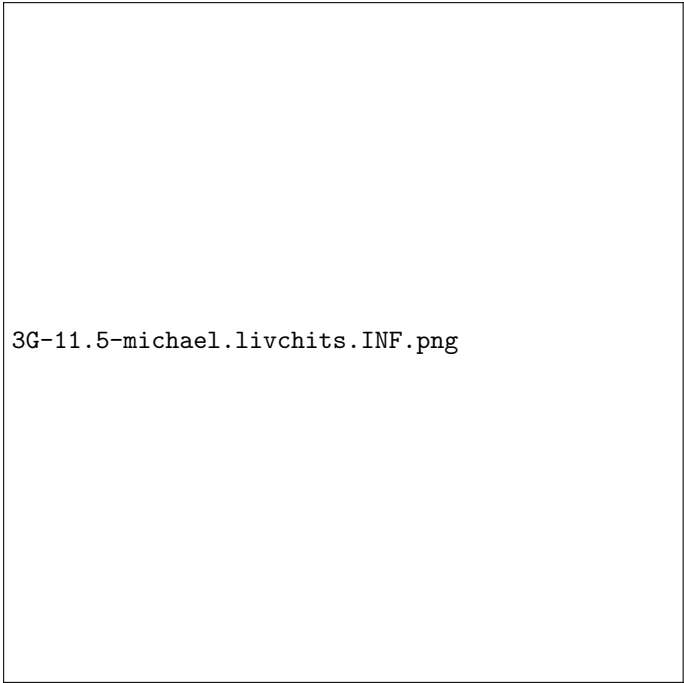
Polen: 0

$$f'(x) = \frac{50}{x^3}$$

Nulpunten: geen

Polen: 0

x	-5		0	5	
$f(x)$	$-$	$-$	$-$	$+$	$+$
$f'(x)$	$+$	0	$-$	$-$	0
$f''(x)$	$-$	$-$	$-$	$+$	$+$
	$\nearrow$ $($	$M(-10)$	$\searrow$ $)$	$\searrow$ $($	$m(10)$ $\nearrow$ $)$



References